

Autopicker Warehouse Robot



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Submitted to

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Objectives

- An “Autopicker Robot” that handles multiple warehouse tasks:
- Picks items directly inside the storage aisles
- Order buffering & sortation
- Stock replenishment
- Brings products to a picking station, where an operator completes the pick
- Used for complex / heavy items, batch picks

Constraints

- Aisle width: must operate in existing aisles
- Floor variations: cope with unevenness ± 10 mm over 1 m and nominal static friction ($\mu \approx 0.3 - 0.6$)
- Temperature/humidity: operate $0 - 40^\circ\text{C}$, 10–85% RH non-condensing
- Shelf types & heights
- Maximum item size: handle items up to $600 \times 400 \times 500$ mm
- Payload: baseline payload 20 kg
- Regulatory: meet local electrical and machine safety regulations
- Power: battery swap or autonomous charging

Design Requirements — Mechanical

Chassis & Drive

- Two drive wheels + 2 casters (omniwheel)
- Nominal footprint: $700\text{ mm} \times 600\text{ mm}$; keep $\text{width} < 800\text{ mm}$ for narrow aisles
- Height: as required
- Top speed: 1.5 m/s (safe walking speed $\approx 1.2 - 1.5\text{ m/s}$)
- Acceleration limit: $\leq 0.5\text{ m/s}^2$ for safe transporting
- Turning radius: zero (if differential) or small for tight aisles

Lift & Manipulator

- Lift mechanism: multi-stage telescopic mast
- Gripper rotation: $\pm 180^\circ$
- Shelf moving in/out: [to define]
- Lift speed: $0.2 - 0.6\text{ m/s}$ (for safe, accurate positioning)
- Payload at gripper: 20 kg continuous
- End effector: flat tray / parallel jaw / vacuum module

Structural

- Materials: welded steel/aluminum subframe; plastic/ABS fairings for covers
- Wheel selection: [to calculate]
- Load analysis: [to perform]